Worksheet No. 8

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1.Aim/Overview of the practical:

To apply the concept of Stored Procedures in database operations to perform tasks like insertion, updating, deletion, and retrieval of data efficiently, securely, and in a reusable manner within the database system.

2. Objectives:

Apply stored procedure concepts for database operations

3. Input/Apparatus Used:

- PostgreSQL
- pgAdmin

4.Procedure/Algorithm/Code:

---CREATING A TABLE:

```
CREATE TABLE Employee (
    empId INT PRIMARY KEY,
    name VARCHAR(50),
    salary INT
);
```

--- INSERT PROCEDURE

```
CREATE OR REPLACE PROCEDURE insert_employee(
    eid INT,
    ename VARCHAR,
    esalary INT
)
LANGUAGE plpgsql
AS $$
BEGIN
```

```
    INSERT INTO Employee VALUES (eid, ename, esalary);
END;
$$;
```

--- UPDATE PROCEDURE

```
CREATE OR REPLACE PROCEDURE update_salary(
    eid INT,
    new_salary INT
)
LANGUAGE plpgsql
AS $$
BEGIN
    UPDATE Employee
    SET salary = new_salary
    WHERE empId = eid;
END;
$$;
```

--- DELETE PROCEDURE

```
CREATE OR REPLACE PROCEDURE delete_employee(
    eid INT
)
LANGUAGE plpgsql
AS $$
BEGIN
    DELETE FROM Employee
    WHERE empId = eid;
END;
$$;
```

--- RETRIVE DATA USING FUNCTION

```
CREATE OR REPLACE FUNCTION get_employees()
RETURNS TABLE(empId INT, name VARCHAR, salary INT)
LANGUAGE plpgsql
AS $$
BEGIN
    RETURN QUERY
    SELECT * FROM Employee;
END;
$$;
```

--- EXECUTION PART

```
CALL insert_employee(1, 'John', 5000);
CALL update_salary(1, 6000);
CALL insert_employee(2, 'Rahul', 5000);

SELECT * FROM get_employees();

CALL delete_employee (1);
```

5. OUTPUT:

	empid integer	name character varying	salary integer
1	2	Rahul	5000
2	1	John	5000

	empid integer	name character varying	salary integer
1	2	Rahul	5000
2	1	John	6000

	empid integer	name character varying	salary integer
1	2	Rahul	5000

7. Learning Outcomes:

1. **Insert Procedure** successfully adds new employee records into the Employee table.
2. **Update Procedure** modifies existing data (e.g., updating salary based on empId).
3. **Delete Procedure** removes specific employee records from the table.
4. **Function (get_employees)** retrieves and displays all employee data in a structured format.

Demonstrates how **procedures handle CRUD operations,** while functions are used for **data retrieval in PostgreSQL**